Date: 10-03-26

Normally we start containers using commands like:

docker run ...

But when an application needs multiple containers
(for example: WordPress + MySQL),

running many docker run commands becomes difficult.

Docker Compose solves this problem by allowing us
to define all containers in one file and start them
with a single command.

Practical 3:
Deploy WordPress + MySql using docker

1. Architecture of the application

   User Browser
   |
   https://localhost:8082
   |
   WordPress Container
   |
   MySql Connection
   |
   MySQL Container
   |
   Database Storage(volumes)

2. Create Project Directory
   Command: `mkdir wordpress-docker`

```
cd wordpress-docker
```

```yaml
services:
 wordpress:
   image: wordpress:latest
   container_name: wordpress_app
   restart: always
   ports:
     - "8082:80"
   environment:
     WORDPRESS_DB_HOST: db
     WORDPRESS_DB_USER: wpuser
     WORDPRESS_DB_PASSWORD: wppassword
     WORDPRESS_DB_NAME: wpdatabase
```

```
    volumes:
      - wordpress_data:/var/www/html
    depends_on:
      - db
  db:
    image: mysql:5.7
    platform: linux/amd64
    container_name: wordpress_db
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: rootpassword
      MYSQL_DATABASE: wpdatabase
      MYSQL_USER: wpuser
      MYSQL_PASSWORD: wppassword
    volumes:
      - db_data:/var/lib/mysql
volumes:
  wordpress_data:
  db_data:
```

```
volumes:
  wordpress_data:
db_data:
```

This is equivalent to the “docker volume create [name]”

Note: I’ve added one extra line in service “db” → “platform: linux/amd64”, because older docker images like ‘mysql:5.7’ have different architecture but I'm using a mac which has arm64.

So the command here say:

platform: linux/amd64 → “Run this container using **AMD64 emulation**, even though the host machine is ARM.”

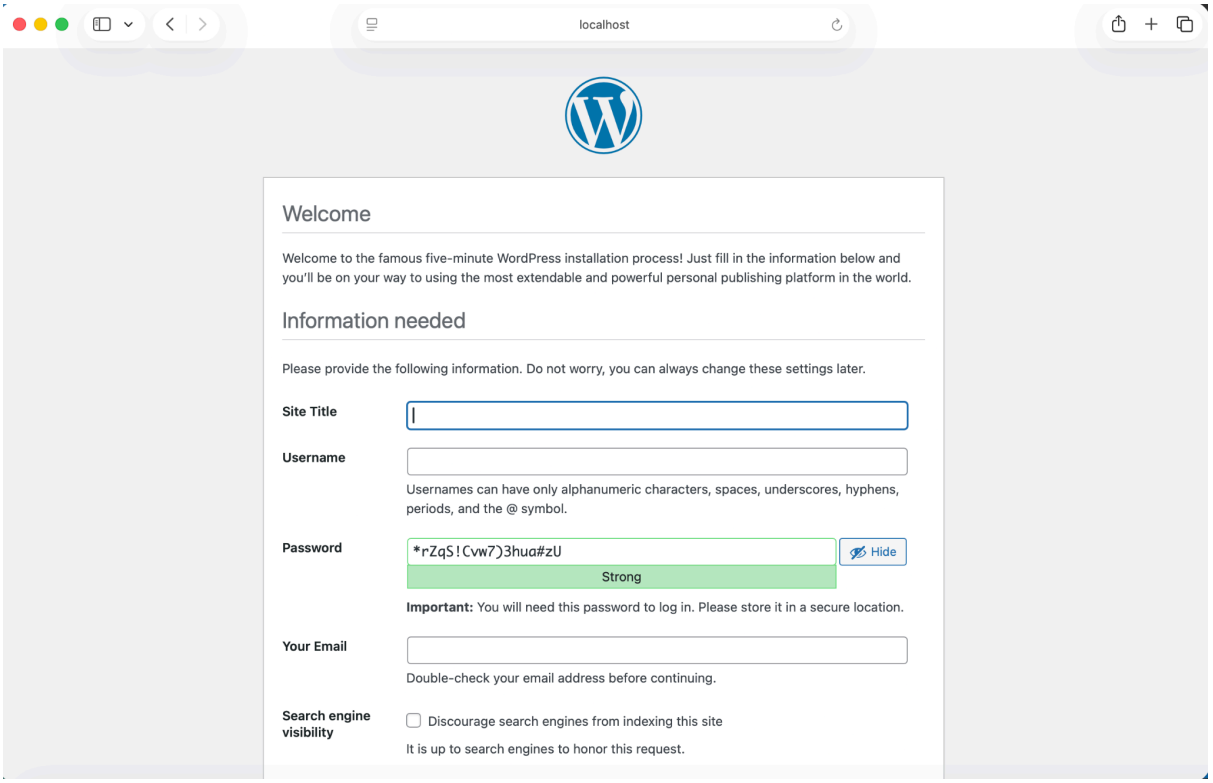
Users > sumitkumarmehta > docker-volume-demo > wordpress-docker

```yaml
Run All Services
services:
  Run Service
  wordpress:
    image: wordpress:latest
    container_name: wordpress_app
    restart: always
    ports:
      - "8082:80"
    environment:
      WORDPRESS_DB_HOST: db
      WORDPRESS_DB_USER: wpuser
      WORDPRESS_DB_PASSWORD: wppassword
      WORDPRESS_DB_NAME: wpdatabase
    volumes:
      - wordpress_data:/var/www/html
    depends_on:
      - db
  Run Service
  db:
    image: mysql:5.7
    platform: linux/amd64
    container_name: wordpress_db
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: rootpassword
      MYSQL_DATABASE: wpdatabase
      MYSQL_USER: wpuser
      MYSQL_PASSWORD: wppassword
    volumes:
      - db_data:/var/lib/mysql
volumes:
  wordpress_data:
  db_data:
```

4. docker compose up -d

Now run command: `docker compose up -d`

```
sumitkumarmehta@sumits-MacBook-Air wordpress-docker % docker compose up -d
[+] up 3/3
 ✔ Network wordpress-docker_default Created
 ✔ Container wordpress_db           Created
 ✔ Container wordpress_app          Created
```

5. Access WordPress

On Browser: https//:localhost:8082